mCOLD Foam V.1 — Service Manual

Version 0.5 (draft) · 2026-10-14 · firmware 0.7.0-dev.8 · hardware foam V1.0.1

For technicians, firmware and QA. Deeper background: `firmware/HANDOFF.md` (state, traps, decisions), `firmware/PROTOCOL.md` (BLE/NFC/MQTT), `firmware/CHANGELOG.md` (per version).

1. Hardware overview

Block	Part	Notes
MCU	ESP32-S3-WROOM-1U-N16R8	16 MB flash, 8 MB octal PSRAM
Power path	BQ25601 charger → TPS63020 → 3V3_MAIN	SW3 switches 3V3_EN; PS/SYNC to GND (power save)
Fuel gauge	MAX17048 (0x36)	Powered from the cell; model not yet tuned to the cell
Current	INA226 (0x40), R15 10 mΩ	Battery branch; 2.5 µV = 0.25 mA per count
USB PD	HUSB238A (0x42)	PD 9 V not enabled
Temperature	MAX6675 + type-K probe	On the e-paper rail (GPIO42); T− tied to GND (rework, see 9)
Display	E-paper, mono panel for now (four-colour panel to come)	SSD1680; rotation 3
Accelerometer	LIS2DW12 (0x18)	INT1 → GPIO2, wakes the box
NFC	ST25DV04KC (0x53/0x57)	GPO → GPIO1
RTC	PCF8523 (0x68) + CR1220 on VBAT_3V	Keeps time with SW3 off
GNSS	ATGM336H on onboard antenna U3	On the SD/GNSS rail (GPIO47)
Lights	3× SK6812MINI-E front, 1× XL-4020 side	LED rail GPIO48, LOW = on
Buzzer	Magnetic, 2.7 kHz, via Q5 (GPIO6)	

Rails, all off at boot: `3V3_EPD_TK` GPIO42 high = on · `3V3_SD_GNSS` GPIO47 high = on · `3V3_LED` GPIO48 low = on.

2. Tools and connection

- PlatformIO (ESP-IDF 5.3.2), project `firmware/`, environment `mcold`.
- USB-C cable to a PC. The box appears as a USB serial port (e.g. COM7), 115200.
- Console: `pio device monitor`. DTR/RTS are held low by `platformio.ini`, so opening the monitor does **not** reset the box. Typed characters are echoed by the box; press Enter to run a command. `help` lists them.
- Only one program can hold the port: close the monitor before flashing.

3. Firmware

Step	Command
Build	`pio run -e mcold`
Flash	`pio run -e mcold -t upload`

Step	Command
Check	boot banner `firmware X.Y.Z` ; console `version`

The version is set in one place: `PROJECT_VER` in `firmware/CMakeLists.txt` . Releases are tagged `firmware/vX.Y.Z` (see `CHANGELOG.md`).

A box asleep on battery has no USB port and cannot be flashed: plug USB in (it wakes and stays awake), then flash. If `sleep_usb` is 1 the box sleeps even on USB — set it back to 0 first.

4. Console commands

Command	What it does
`help`	List of commands
`health`	Every device, its state and readings; battery, charger, rails
`tasks`	Task heartbeats, heap
`version`	Firmware version and image hash
`sn` · **`sn set S`**	The serial number; factory sets it, then reboot (screen, BLE, NFC, MQTT, records)
`accel` / **`gnss`** / **`nfc`**	Sensor detail
`led I R G B` · **`ledtest [S] [N]`** · **`beep MS`**	Lights and buzzer by hand
`config` · **`config set K V`** · **`config reset`**	Settings (section 5)
`trip` · **`trip start [L H [HYST DWELL]]`** · **`trip stop`** · **`trip ack`** · **`trip dump [N]`**	Trips from the bench
`trip csv [ID]`	A trip as its CSV file (`TRIP_<SN>_<YYMMDDhhmm>.csv`)
`log` · **`logtest`**	Trip log status; power-cut self-test (14 cases)
`screen` · **`screen N`** · **`screen rot 1|3`**	Redraw, design pages, orientation
`screen cal`	Six frames 3 px apart: count those visible per side, inset = (6 – count) × 3
`wifi` / **`sync`**	Wi-Fi and upload status
`wifi add SSID PASS` · **`wifi del SSID`**	Known networks (up to 5)
`wifi ip SSID dhcp` · **`wifi ip SSID IP GATEWAY SUBNET [DNS]`**	A network's address: DHCP or fixed
`mqtt set HOST PORT [USER PASS]`	Broker and login (the console is the only way to set it)
`ota` · **`ota base URL`** · **`ota FILE|URL [force]`**	Firmware update state, file server, install now
`flash`	Restart into download mode, to flash over USB (or run `python tools/usb_flash.py`)
`ble [on|off]`	BLE status; `on` advertises 60 s
`rpc {json}`	Any PROTOCOL.md request, authorized
`sleep` · **`sleep clear`**	Power manager state, wake record

Command	What it does
`soc` · `soc set P`	Counted state of charge, capacity, anchor; set it by hand
`reboot`	Restart

Bench-only (power measurement and tests):

Command	What it does
`amps [S]`	Battery current for S s: mean, min, max
`hiz [S]` · `hiz off`	Charger input off for S s: the box runs on its cell with USB in. Every boot switches it back on
`sleep test S`	Deep-sleep S s, timer wake only (even on USB)
`poweroff`	The battery-empty path: screen, ship mode. Unplug and replug USB to restart
`cpu MAX MIN LS`	CPU clock / light sleep until the next boot
`pin N 0|1|in`	Take a pin over as plain GPIO (until reboot)
`log write N` · `log read` · `log erase`	Bench trip records
`ack {json}`	Inject a server ACK
`cfgdoc {json}`	Inject a server settings document (`docs/app/device-settings.md`)
`i2c`	SDA/SCL levels and every I2C address that answers
`trip fill N`	Test only: N sample rows at once (a long trip for the USB drive)
`ota rollback-test`	The next new image will not confirm itself
`config set usb_drive 0`	Turn the USB drive off: the box restarts with the port as before
`config set tap_test 1`	Test only, never in the product: a double tap is a command, by how the box lies -- normal: start/stop a trip (1/2 beeps), left side: redraw the screen (3), right side: GNSS for 10 min (4); one long beep: not a command. Motion is not counted while on
`config set rx_show 1`	Test only, never in the product: the screen shows what a phone sent (NFC tap, BLE connect, each command and its answer; secrets masked) for a minute after the last
`tap ths N`	With tap_test: the double-tap threshold, 1..31 x 62.5 mg (9 = 562 mg), until the next boot
(USB drive)	`DEVICE.TXT` + one `TRIP_<SN>_<YYMMDDhhmm>.csv` per trip, read-only, a snapshot from when the cable went in
`hang`	Hang the console task: the supervisor restarts the box after 5 min

5. Settings

Stored in NVS, kept through firmware updates. `config` shows each with its range and whether it is the default.

Key	Range	Default	Meaning
`sample_period_s`	60–3600	300	Trip sample period
`cal_offset_c100`	−1000–1000	0	Temperature offset, 0.01 °C
`cal_gain_ppm`	900000–1100000	1000000	Temperature gain
`cal_version` / `cal_date`	—	0	Calibration record
`temp_adj_c100`	−1000–1000	0	User adjustment added after the calibration, 0.01 °C (eTEMP's tempAdj); settable from the app and the server
`accel_wake_ths`	1–63	2	Motion threshold, ×31 mg
`led_bright_pct`	1–100	20	Side light brightness cap
`led_front_pct`	1–100	2	Front lights brightness cap
`led_status_s`	60–3600	900	On battery, the alive blink this often
`batt_trip_mv`	3000–4100	3550	Below it, on battery, no new trip (`BATTERY_LOW`)
`batt_off_mv`	3000–3700	3400	Below it, on battery, the box switches itself off
`batt_mah`	300–6000	1500	Rated cell capacity, until one is learned
`sleep_ua`	0–5000	250	Current asleep, for counting charge through a sleep
`soc_source`	0–1	0	Percent shown: 0 voltage (OCV table), 1 counted
`tz_offset_min`	−720–840	420	Display time zone (records are UTC)
`epd_inset_t/b/l/r`	0–20 / 0–30	0	Panel pixels the case hides on each side (`screen cal`)
`buzzer_enabled`	0–1	1	Alarm sound
`buzzer_volume`	1–100	100	Alarm sound loudness, % of full drive; settable from the app and the server
`alarm_dwell_s`	0–3600	300	Seconds outside the limits before an alarm is raised; settable from the app and the server, locked during a trip
`alarm_hyst_c10`	0–100	5	An alarm clears this far inside the limit, 0.1 °C; settable from the app and the server, locked during a trip
`chg_in_ma`	100–3200	400	Charger input limit (IINDPM), mA: the most the box draws from USB. 400 mA × 6 boxes × 5 V = 12 W, for the dock's 15 W adapter. The box cannot tell the dock from another USB source, so every source gets it; raise it for a single box on a bench

Key	Range	Default	Meaning
`chg_ma`	60–3000	300	Charge current into the cell (ICHG), mA, 60 mA steps. Set again whenever the charger has put its own back (every plug-in)
`uplink_en`	0–1	1	Bench: 0 keeps the box off the broker (a retained `online` "0" is sent first)
`sleep_en`	0–1	1	Deep sleep on battery
`idle_wake_s`	300–86400	3600	Longest sleep with nothing due
`gnss_period_s`	300–86400	1800	On battery in a trip: one GNSS session this often
`upload_period_s`	60–86400	300	On battery: one upload session this often
`sleep_usb`	0–1	0	Bench: behave as on battery with USB in
`sleep_meas`	0–1	0	Bench: measure each sleep's current (+0.33 mA)
`light_sleep`	0–1	0	Trial: light sleep between tasks while awake, on battery

Wi-Fi and broker credentials are separate (`wifi add`, `mqtt set`) and never appear in documents or the repository. Which keys the app and the server may set, and the settings document they use, are in `docs/app/device-settings.md`; the calibration (`cal_*`), the battery limits and the bench switches are the console's only.

6. Provisioning a new box

1. Flash the current release; `sn set <label SN>`, then `reboot`.
2. `health`: every fitted device `ok` (pd, buzzer, sd show `unknown` until used).
3. `logtest`: all 14 pass.
4. `ledtest` and `beep 150`: check each light and the buzzer by eye and ear.
5. `wifi add …` for the site networks; `mqtt set …` for the broker.
6. `sync`: Wi-Fi connected, broker connected.
7. Check the time: `health` → clock source `ntp`, `gnss` or `rtc`.

7. Power

Measured (bench, 2026-10-03)	Value
Asleep	≤ 0.25 mA (at the current monitor's floor)
Awake, radios off	~130 mA (under investigation, see 10)
Awake, BLE stack on	~155 mA
Awake, GNSS on	~173 mA
Charging from a PC port	+330 to +400 mA into the cell
Timer wake	0.13 s boot + 1.2 s awake; 3.6 s with a refresh

Behaviour on battery: deep sleep between jobs; wakes on the sample timer, motion, an NFC tap or USB power. USB power keeps the box awake.

Low battery: below `batt_trip_mv` no trip starts; below `batt_off_mv` (three readings in a row) the box records `POWER OFF` in the trip, shows BATTERY EMPTY and puts the charger in ship mode — the cell is disconnected from everything but the fuel gauge. Plugging USB in powers it back on (verified 2026-10-03).

State of charge is counted, not read from the voltage: the current monitor's readings while awake, `sleep_ua` for each sleep, 100 % when the charger reports charge done; the fuel gauge's percent only to start, or when the kept count is more than 15 % from it (charged while switched off). A run from full to the switch-off voltage teaches it the cell's real capacity. `health` shows the counted and the gauge's percent side by side. The switch-off itself is by voltage.

Wake record: `sleep` lists recent wakes (cause, boot ms, awake ms, mA, slept s). It survives a reset but not a power cut; `sleep clear` restarts it.

8. Diagnostics

`health` state	Meaning
`ok`	Answering
`degraded`	Worked before, failing now: something came loose
`failed`	Never worked this boot: not fitted or built wrong
`unknown`	Not tried yet

Failed devices are retried once a minute and recover on their own.

Trip records (`trip dump N`): SAMPLE (temperature, motion, GNSS, battery, alarms), EVENT (RESUMED, MOTION, PROBE FAULT/OK, ALARM, ALARM CLEAR, ALARM ACK, TIME SET, LOSS, POWER OFF), START, STOP.

9. Known hardware rework (every board of this BOM)

Item	Do
GNSS RF path	Leave off C_M1, C_M2 and R_SET_IPEX (onboard antenna). For the u.FL instead: off R_SET_PCB, C_M1, C_M2; keep R_SET_IPEX
RTC backup	CR1220 to VBAT_3V (pad hand-soldered on V1.0.1)
Thermocouple	MAX6675 T– to GND, or an open probe is not detected
Buzzer	Its pulse sets off the accelerometer; firmware ignores motion during a beep

10. Troubleshooting

Symptom	Check
No USB port	Box asleep on battery: plug in USB. Still none: SW3, then the battery
`pio` upload fails, port busy	Close the serial monitor
Box dark after `poweroff` or an empty battery	Unplug and replug USB. If not: disconnect and reconnect the battery
Temperature `--`	Probe plugged in? `health` → thermo

Symptom	Check
Clock `----------`	No source yet: Wi-Fi (NTP), sky (GNSS), or the app; check the CR1220
No GNSS fix	Indoors is normal; sessions give up after 60 s with nothing heard
Records never leave the box	Server not sending ACKs (`sync` shows pending)
Battery % over 100 / "no battery"	No cell connected: the gauge sees the charger's output
Motion events with nobody near	Under investigation; buzzer and rail switching are already blanked
Awake current ~130 mA	Under investigation (bench floor ~40 mA)

Not yet included

- Battery: cell datasheet, charge current and limits, fuel-gauge model, runtime
- Root cause of the ~130 mA awake current
- Charger / PD policy, HUSB238A register map, PD 9 V
- USB drive: Windows checks (phase F4); SD archive (deferred)
- Dock (6 boxes), factory provisioning tool, production logging, secure boot / flash encryption (P8)
- Temperature calibration procedure and accuracy
- Four-colour display, shock alarm, door sensor
- Disassembly, part replacement, schematics references by page